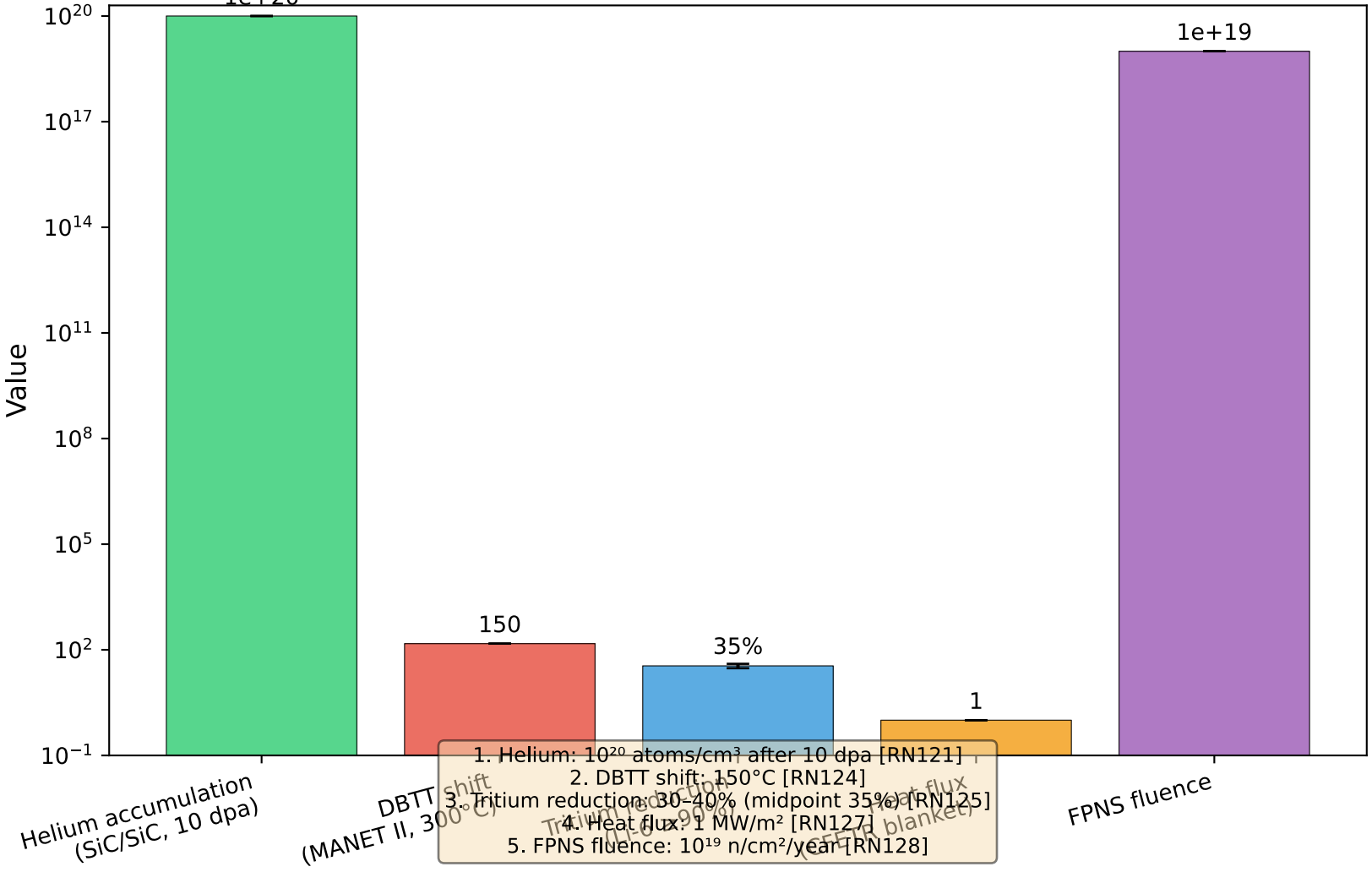

Key Quantitative Metrics from Ceramic Breeder Development Gaps
Data extracted from Section 5: Gaps and Recommendations
Value
1e+20
150
35%
1
1e+19
Helium accumulation (SiC/SiC, 10 dpa)
DBTT shift (MANET II, 300°C)
Tritium reduction (30–40%)
Heat flux (CFETR blanket)
FPNS fluence
1. Helium: 10²⁰ atoms/cm³ after 10 dpa [RN121]
2. DBTT shift: 150°C [RN124]
3. Tritium reduction: 30–40% (midpoint 35%) [RN125]
4. Heat flux: 1 MW/m² [RN127]
5. FPNS fluence: 10¹⁹ n/cm²/year [RN128]